

The Upper Limb



Dr. Rasha Alshali

Bones of the upper limb

Includes 32 bones:

Shoulder girdle:

Clavicle, anteriorly
Scapula, posteriorly

Upper arm:

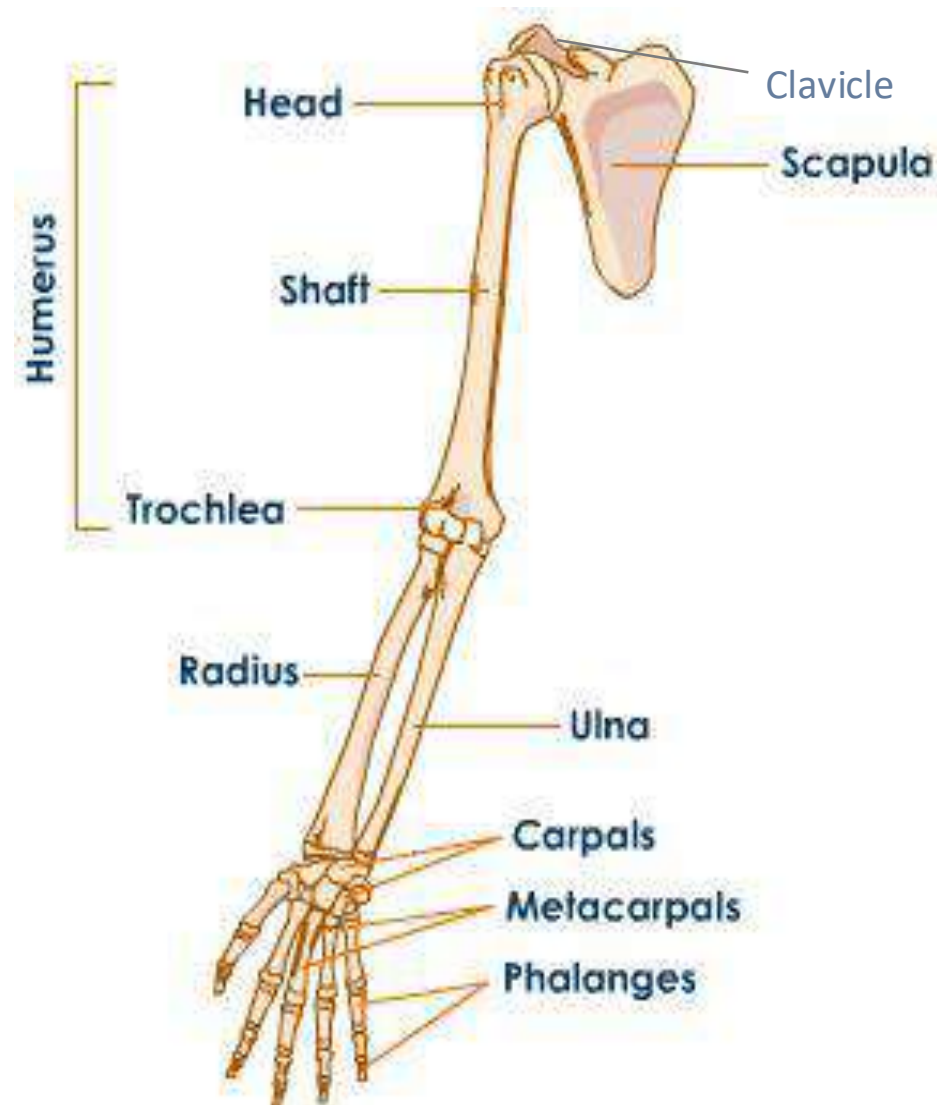
Humerus

Forearm:

Radius, laterally
Ulna, medially

Hand:

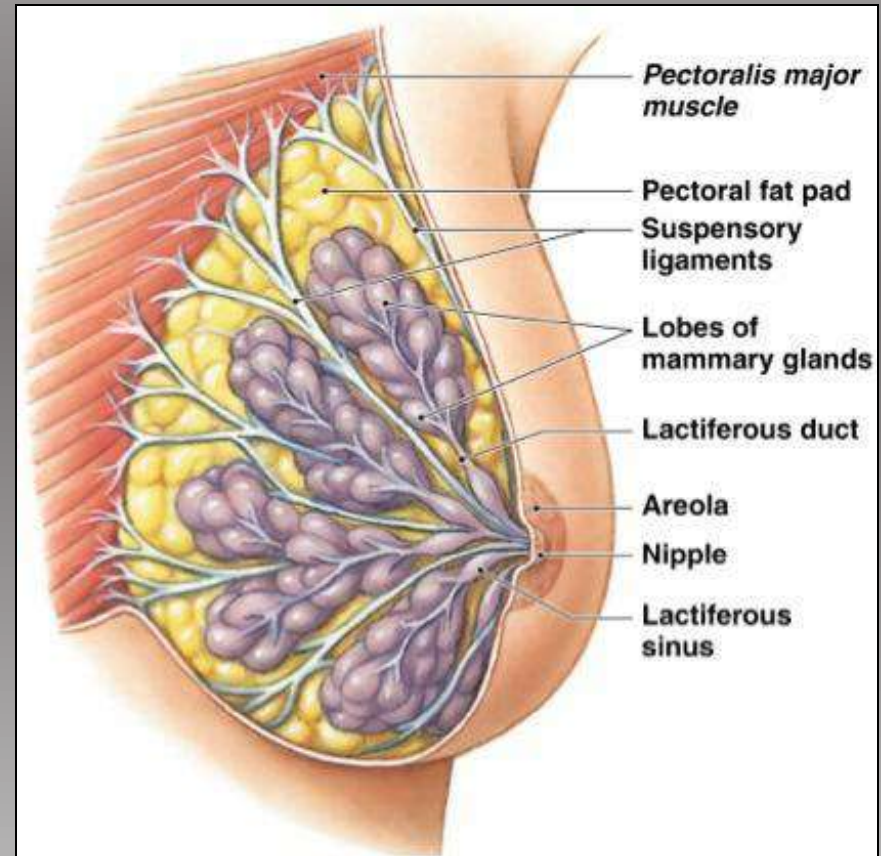
Carpals (8 bones)
Metacarpals (5 bones)
Phalanges (3 in each finger,
except the thumb only 2 phalanges)



Pectoral Region

And

Mammary Gland



Pectoral Region



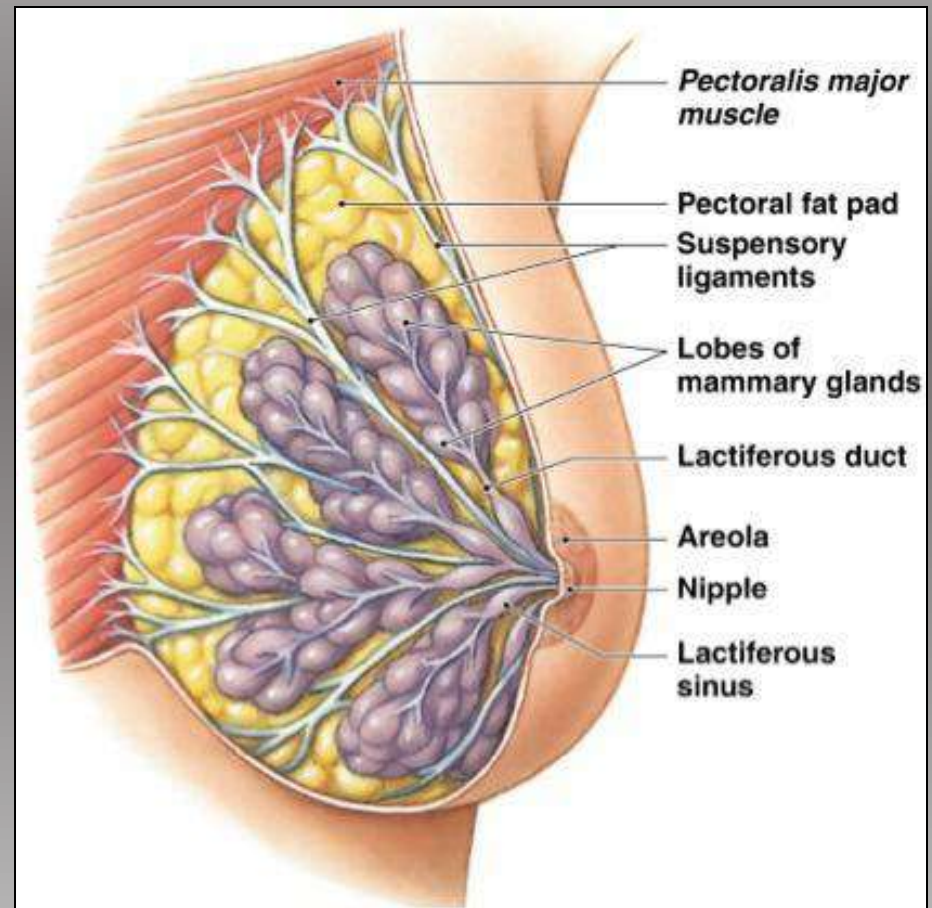
The pectoral region is the region which connects the upper arm to the front of the chest.

- Bones
- Cutaneous nerves
- Muscles, nerves, vessels & lymphatics
- **Clavipectoral fascia**

Mammary Gland

Specialized accessory glands of the skin which can secrete milk.

- Site, description,
- Blood supply & lymph drainage of the breast.
- **Breast Carcinoma:**
LN involved,
Clinical features.
- **Breast abscess:**
causes & drainage



The pectoral region

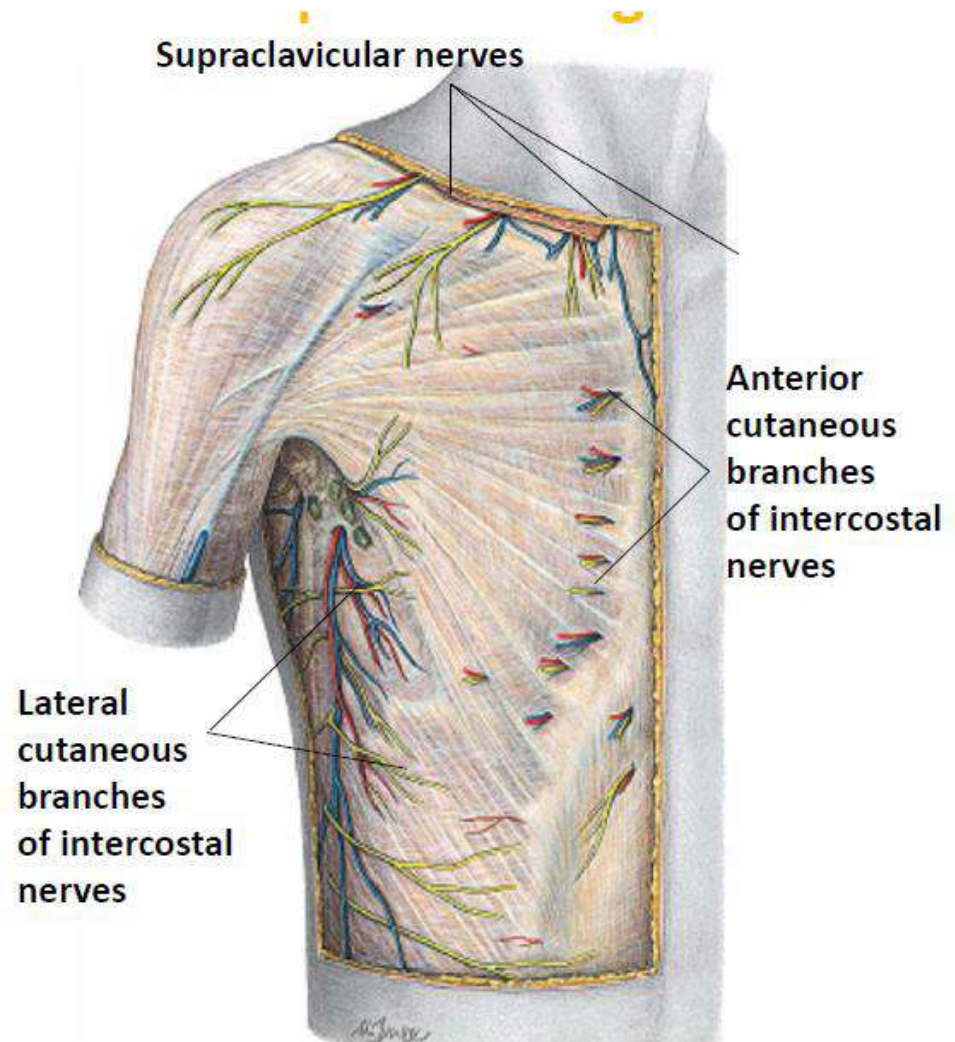
Cutaneous nerves:

The **superior** pectoral region:

1. Supraclavicular nerves C3 & C4

The **inferior** pectoral region

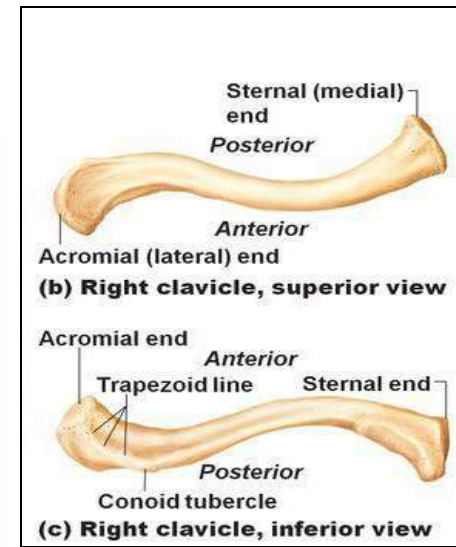
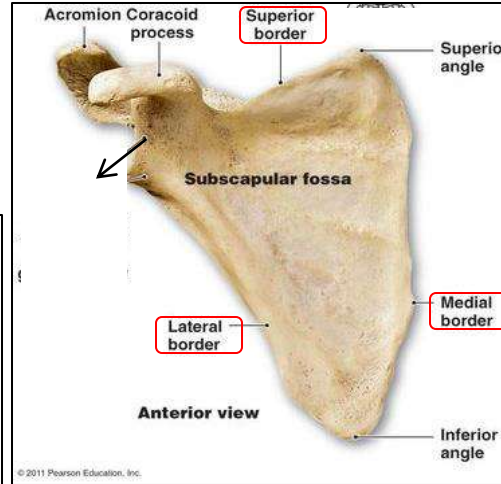
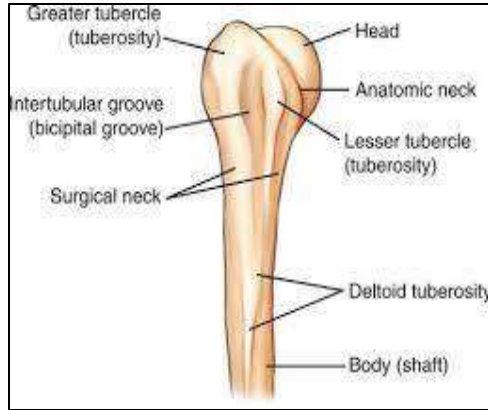
2. Anterior & lateral cutaneous branches of intercostal nerves (T3 , T4 & T5)



Bones:

Pectoral girdle

The Clavicle. The Scapula.



The Humerus.

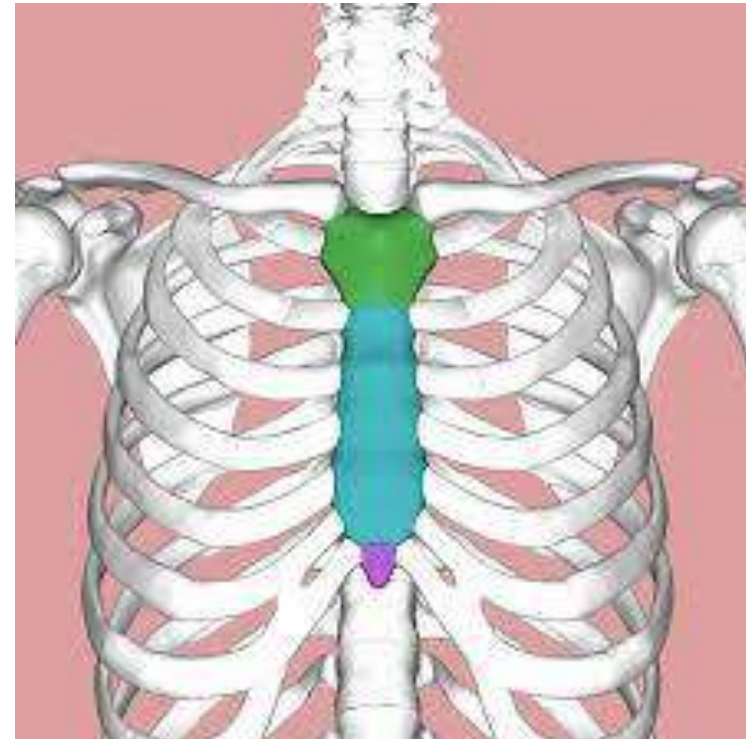
The Sternum:

lies in the midline of 'anterior chest wall.
It is a flat bone, divided into three parts.

The Ribs:

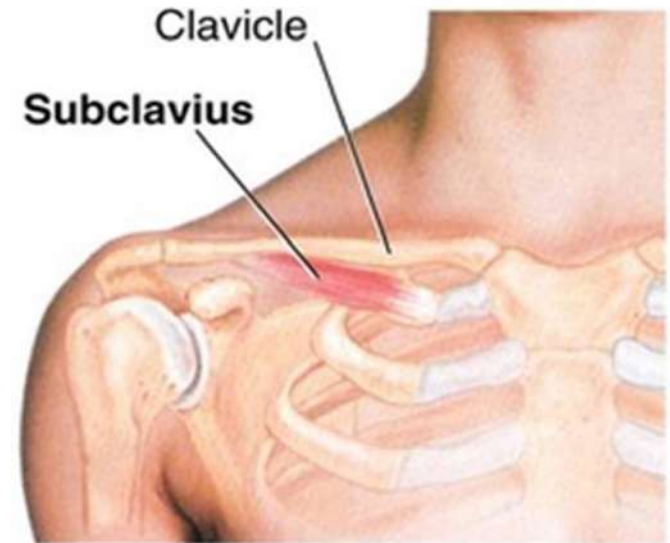
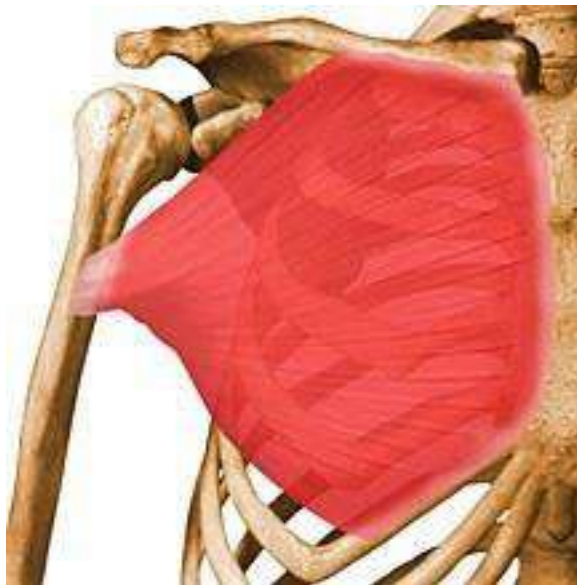
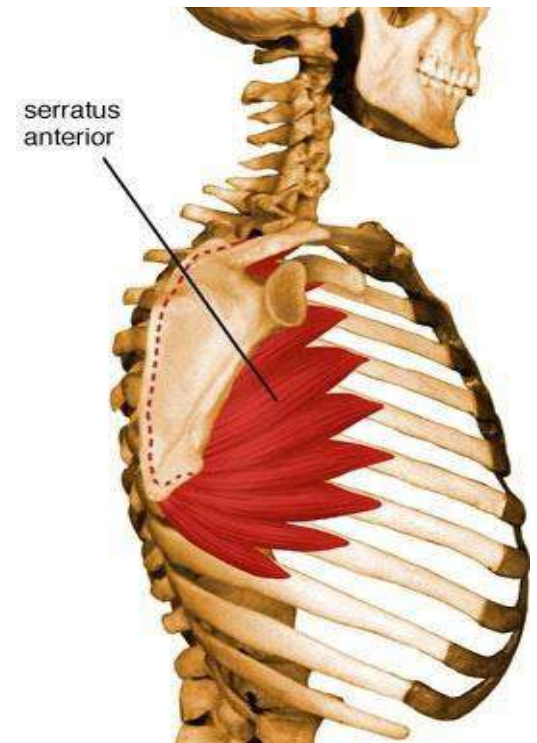
12 pairs of ribs, attached:

- posteriorly to 'thoracic vertebrae
- anteriorly to 'sternum through costal cartilage.



Muscles :

- ☐ Pectoralis major
- ☐ Pectoralis minor
- ☐ Subclavius
- ☐ Sarratus anterior



Pectoralis major

A large thick triangular muscle.

- **Origin:** medial half of **clavicle**, **sternum**, upper six **costal cartilages**.

- **Insertion:** Lateral lip of bicipital groove of **humerus**.

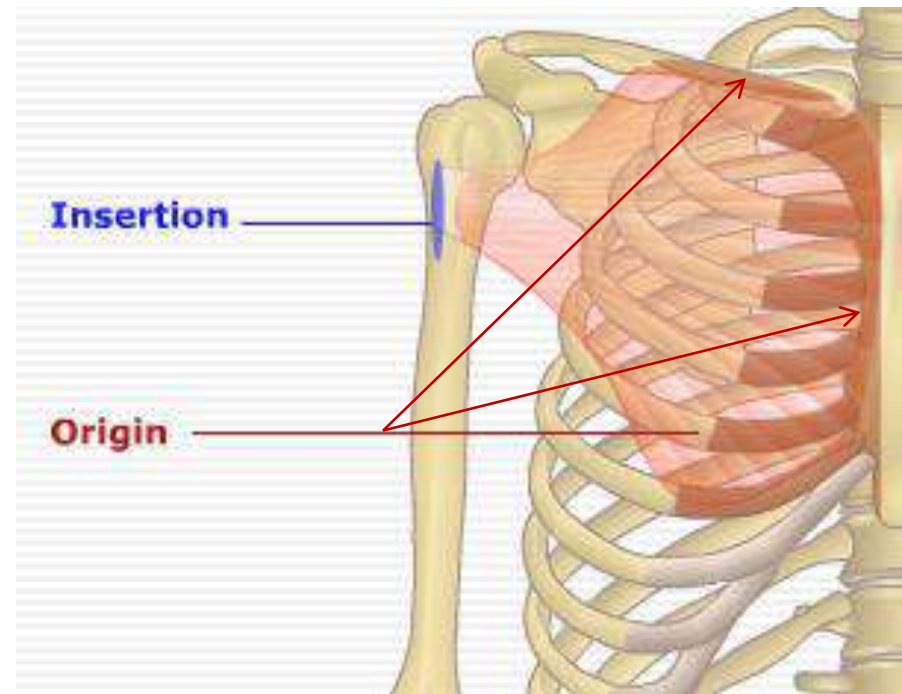
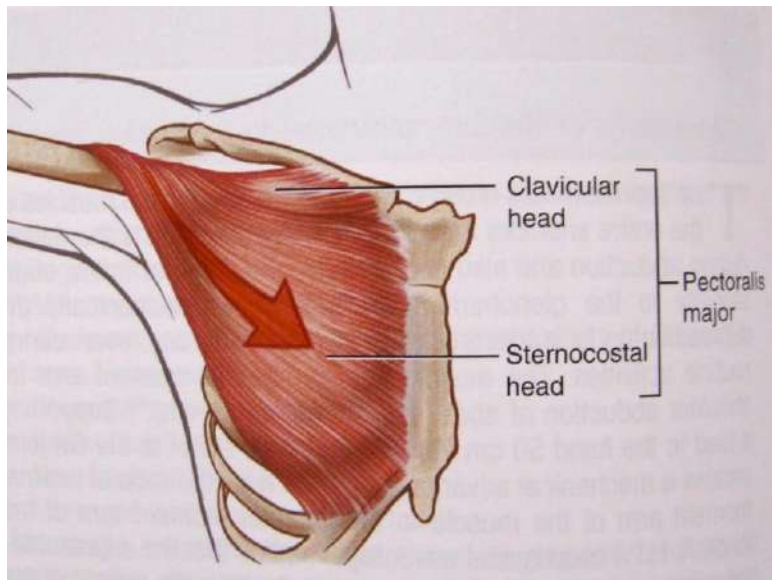
- **Nerve supply:**

Medial & Lateral pectoral nerves

- **Action:**

Adduction & Medial rotation of arm

Clavicular fibers → arm flexion
at shoulder joint



Pectoralis minor

A thin triangular muscle that lies beneath the pectoralis major.

- **Origin:** from 3rd, 4th, and 5th **ribs** and runs upward & laterally
- **Insertion:** by its apex into 'coracoid process of 'scapula.

- **Nerve supply:**
Medial pectoral nerve

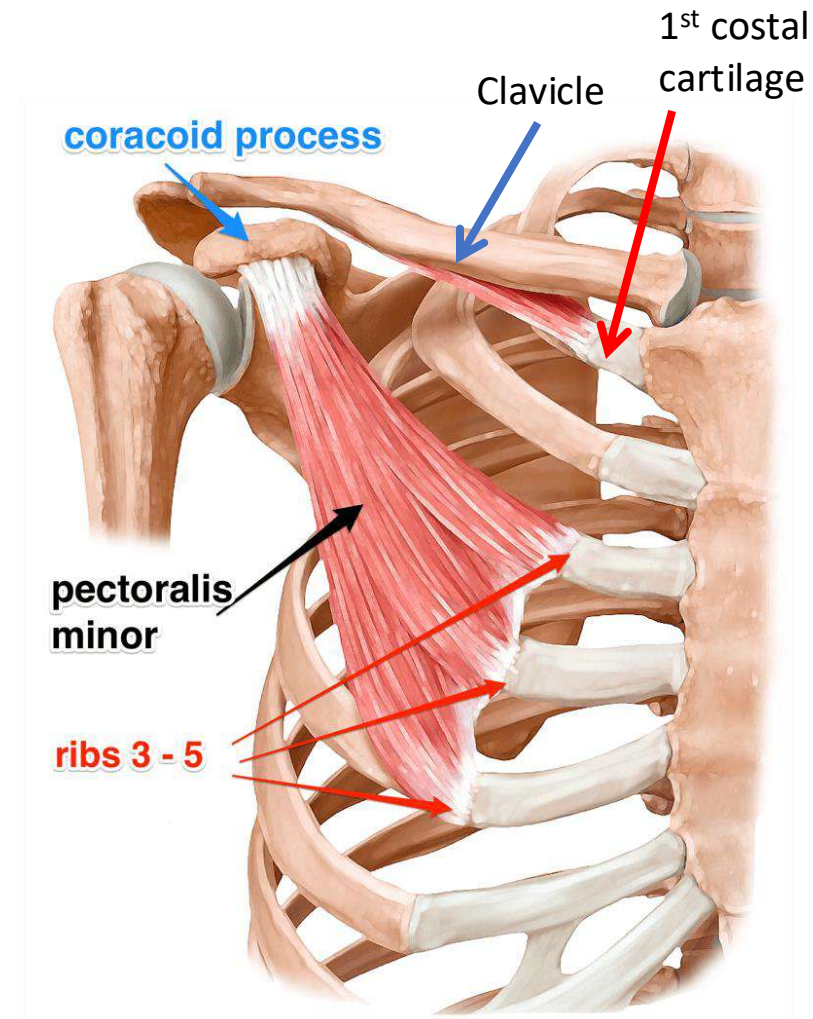
- **Action:**
Depresses the point of shoulder;
if the scapula is fixed →
it elevates the ribs of origin (during inspiration)

Subclavius

- **Origin:** from 1st **costal cartilage**
- **Insertion:** undersurface of the **clavicle**.

- **Nerve supply:**
Nerve to subclavius

- **Action:**
Depresses the clavicle.



Sarratus anterior

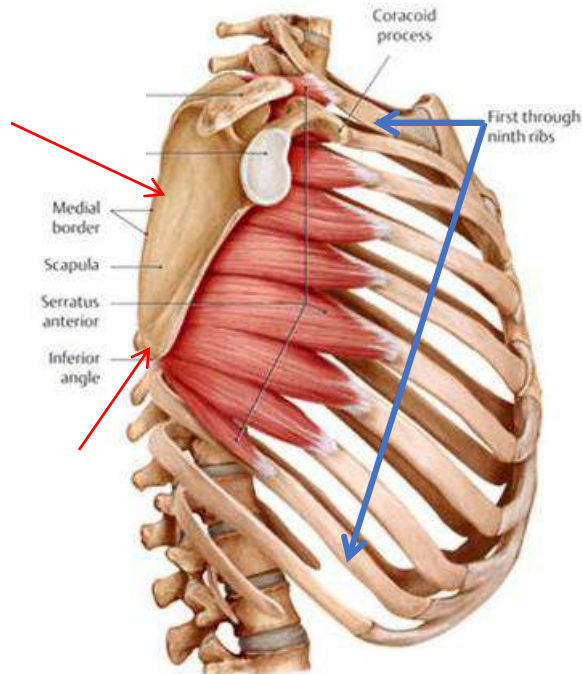
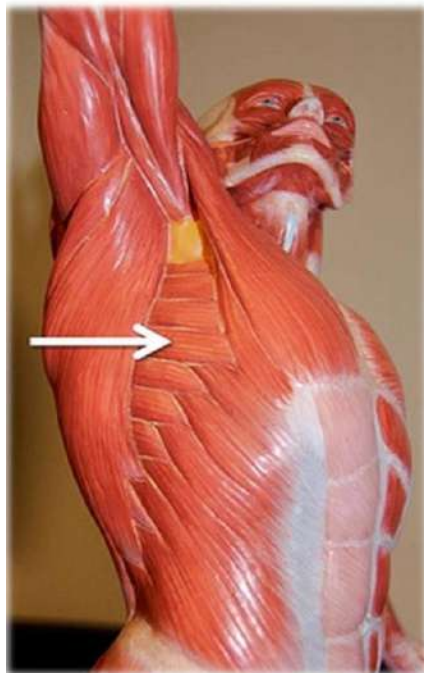
It is a large thin muscle that covers the lateral chest wall.

- **Origin:** from Upper eight [ribs](#)
- **Insertion:** anterior aspect of medial border & inferior angle of [scapula](#).
- **Nerve supply:** Long thoracic nerve
- **Action:**

Draws 'scapula **forward** around 'thoracic wall; (protraction of scapula)

Rotates 'scapula (inferior angle passes forward & laterally)

Strong accessory muscle of inspiration



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Winged scapula

“injury of nerve to serratus anterior”



Clavipectoral Fascia

It is a thickened fascia filling the gap between **subclavius** & **pectoralis minor** muscles.

Superiorly: it splits to enclose **subclavius**, then attached to 'clavicle.

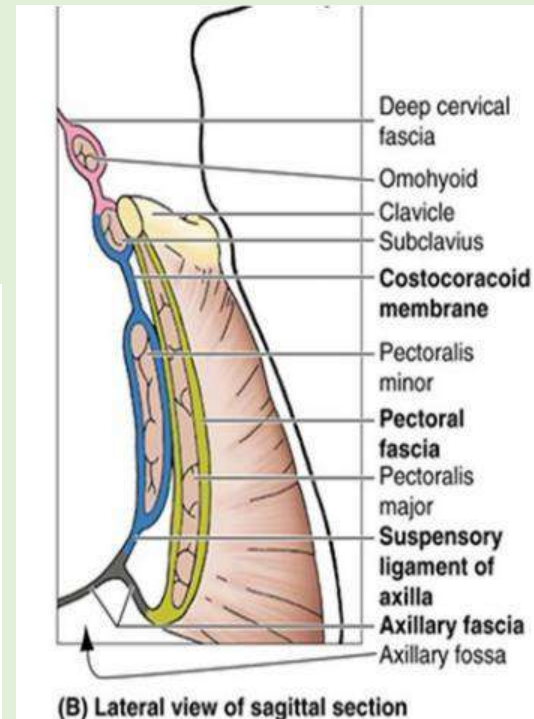
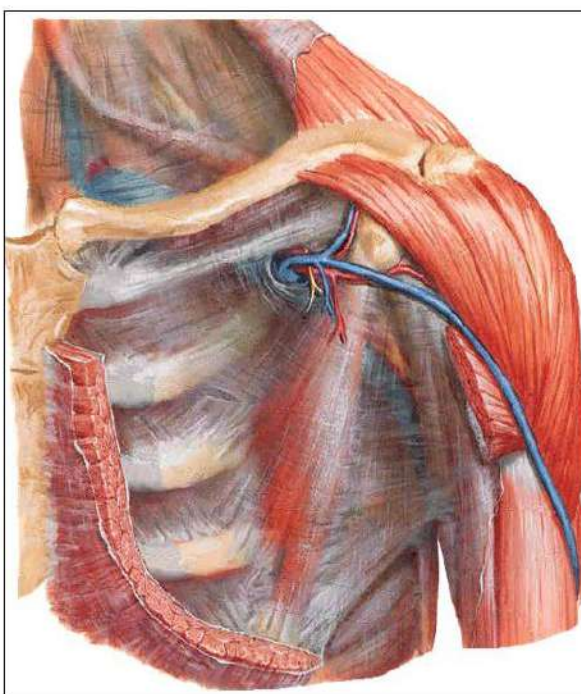
Inferiorly: it splits to enclose **pectoralis minor**, then extends down as the **suspensory ligament of axilla** which is attached to axillary fascia

Medially: attached to 1st, 2nd costal cartilages.

Laterally: attached to coracoid process.

Structures Piercing the Clavipectoral Fascia:

- ✓ Thoraco-acromial **artery**.
- ✓ Cephalic **vein**.
- ✓ Lateral pectoral **nerve**.
- ✓ **Lymphatic vessels**.



The Breasts (Mammary Gland)

- The breasts are specialized accessory glands of the skin which can secrete milk.

Shape: it is conical or hemispherical.

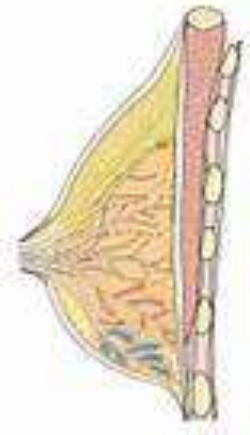
-Left breast is usually slightly larger.

- ✓ It is fully developed in adult females.
- ✓ **It is rudimentary:** In female before puberty &

Male breast



Female breast

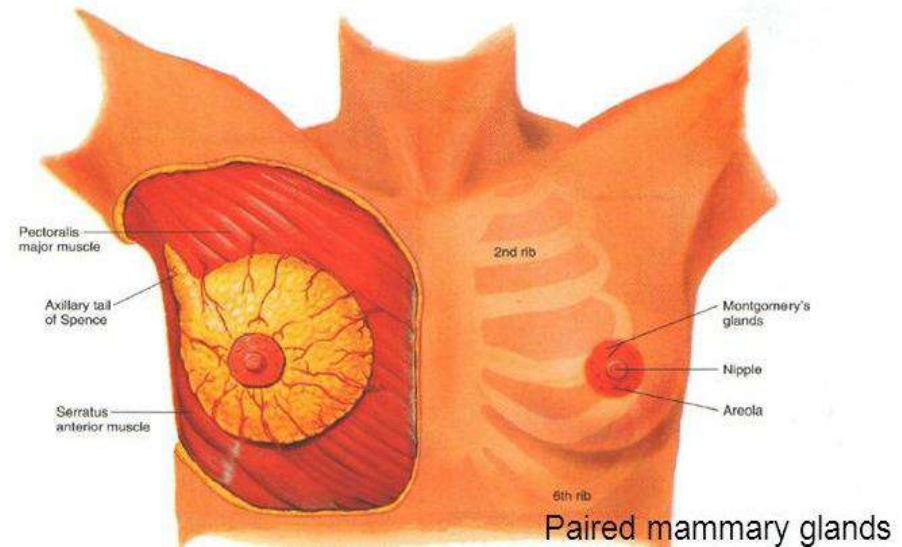


Location Of The Breast

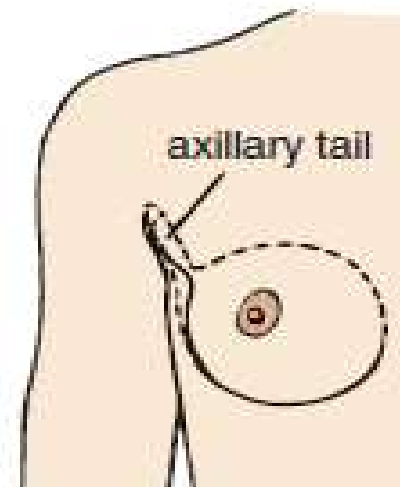
In lies in the **superficial fascia** of the

❖ **Base of the breast extends:**

- **Superiorly:** 2nd rib.
- **Inferiorly:** 6th rib.
- **Medially:**
lateral margin of the sternum.
- **Laterally:**
Mid-axillary line



- **Axillary tail** extends upward & laterally pierces deep fascia & enters the axilla.



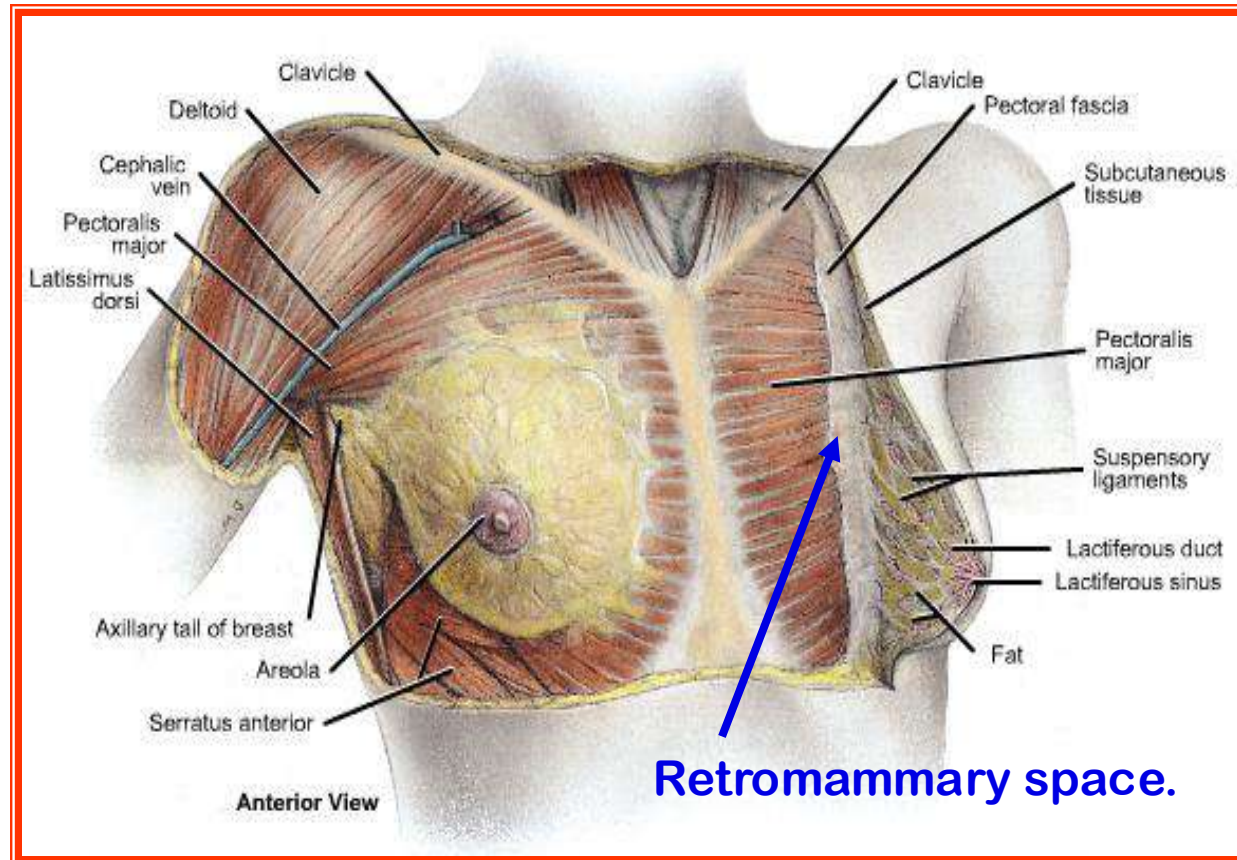
Relations:

supero-medial 2/3:

rest on deep fascia covering **pectoralis major** muscle
(separated from it by **retro-mammary space**).

infero-lateral 1/3:

rests on **serratus anterior** & **external oblique** muscles



External Structure Of The Breast

The skin of the breast shows:

➤ The nipple:

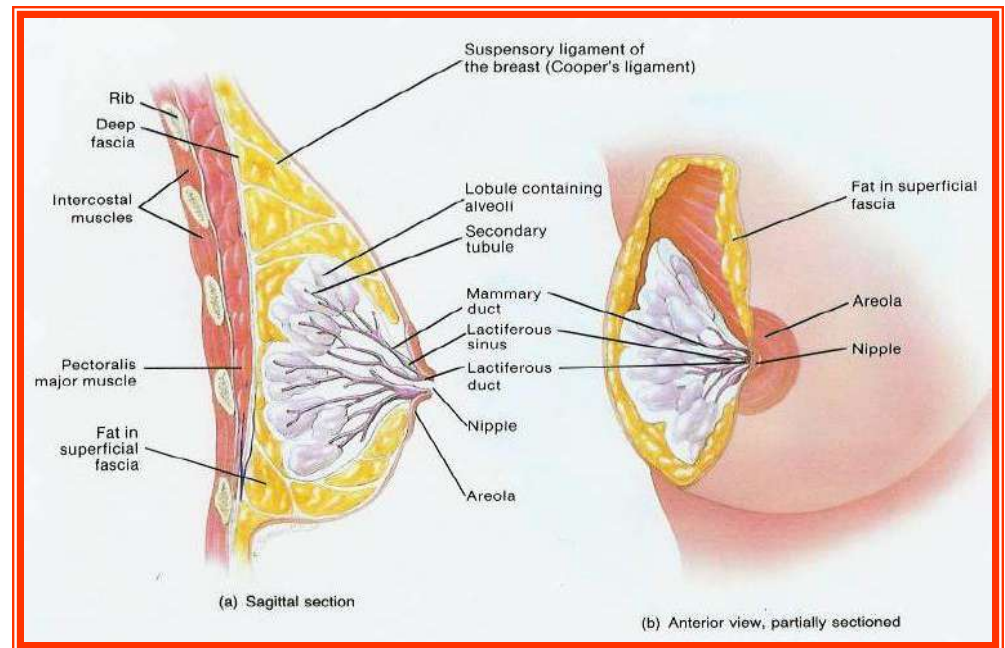
A conical projection lying opposite to the *4th intercostal space* in males and in females before pregnancy.

➤ The areola:

area of pigmented skin surrounding the nipple, lacking of hair. It is pink in nulliparis but change to dark brown in pregnancy.

➤ There is no subcutaneous fat beneath the nipple and areola.

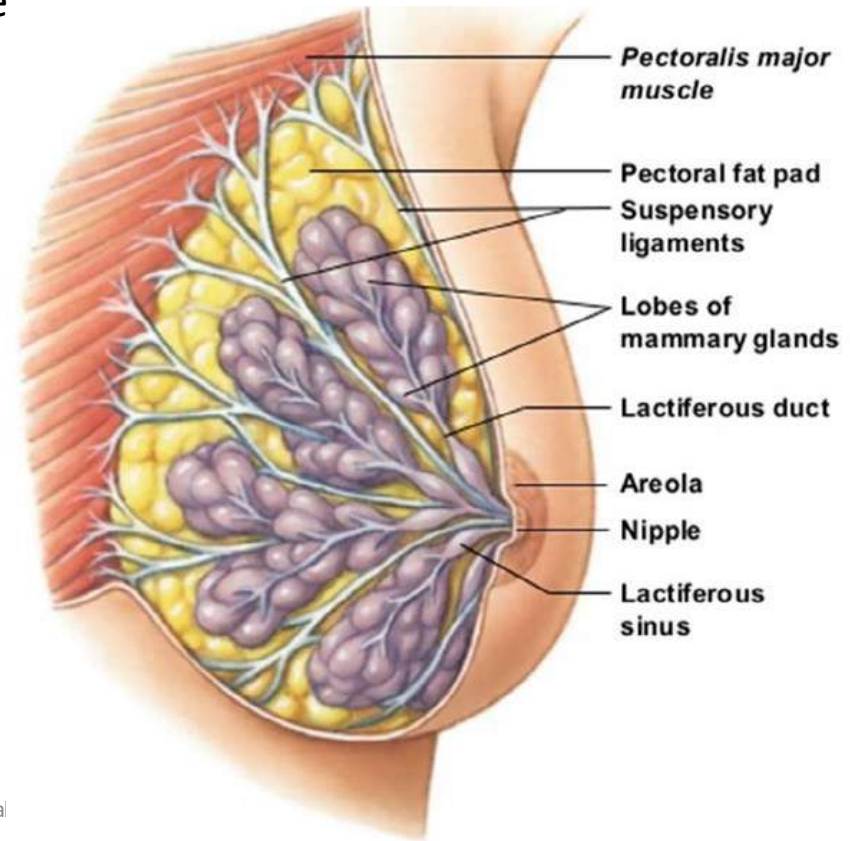
➤ The breast has No capsule.



Internal Structure Of The Breast

- ❑ It is formed of **lobes / lobules** and **duct system** embedded in **adipose** (fatty) tissue.
- ❑ Each breast consists of **15 -20 lobes**, each lobe is divided into **lobules** radiating out from the nipple.
- ❑ From each lobe → main duct called **mammary/ lactiferous ducts** (15-20) converging toward 'nipple & dilate to form **lactiferous sinus**.
Then, opens separately on the summit of the nipple

- The lobes are separated by **fibrous septa**, which extend from the skin to the pectoral fascia (**suspensory ligament of Cooper**).
- Behind the breast is a space filled by loose connective tissue called **the retromammary space**.



Blood Supply Of The Breast

The arterial supply:

❑ **Medially:** 60% of the blood supply:

1- A branch from **subclavian artery** → **internal thoracic (mammary) artery**..

Giving **Perforating** branches → **medial mammary arteries**.

❑ **Laterally:**

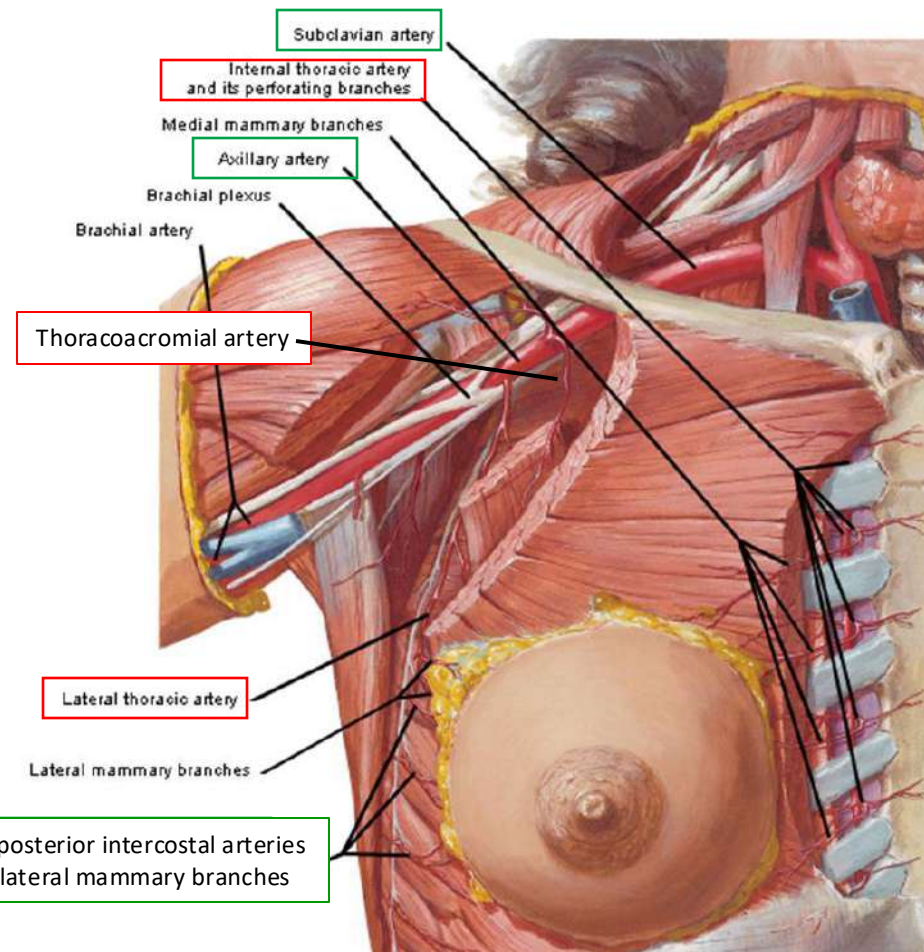
2- Branches of **axillary artery**:

- The **thoracoacromial** artery
- The **lateral thoracic** artery → **lateral mammary arteries**.

3- The **posterior intercostal arteries** → **lateral mammary arteries**.

The Venous drainage

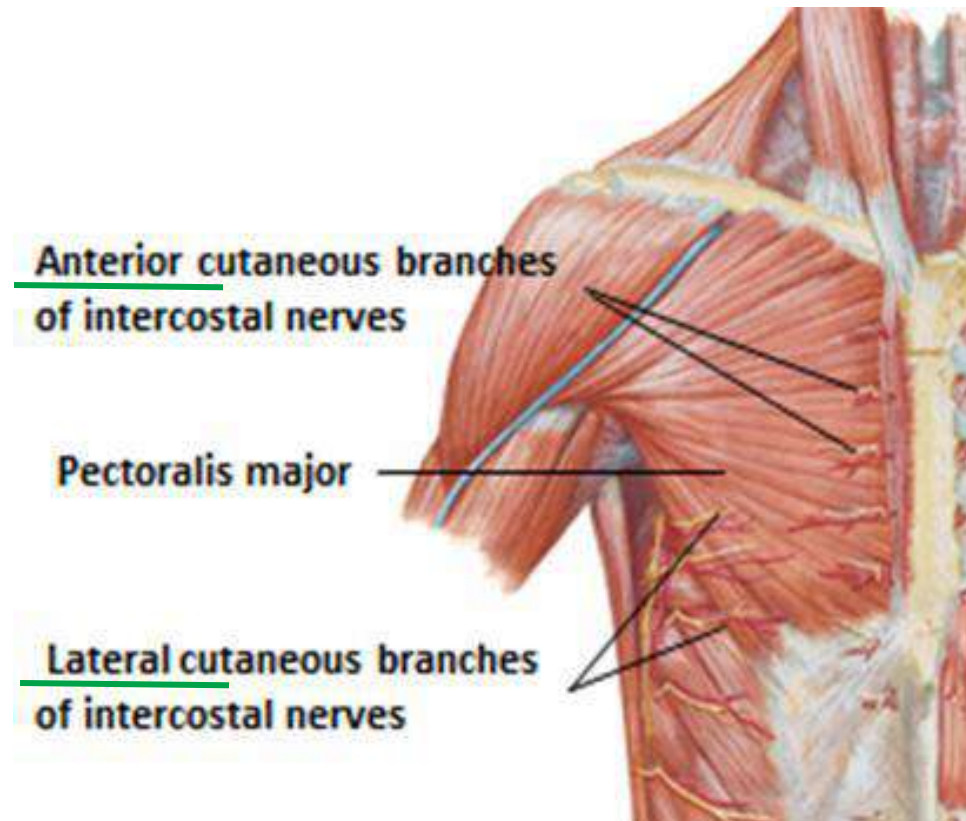
correspond to 'arterial supply drains into:
(internal thoracic, axillary veins &
Posterior intercostal v.)



Nerve Supply Of The Breast

The 4th , 5th , 6th intercostal nerves give →
anterior & lateral cutaneous branches to innervate the breast.

These nerves contain both sensory & autonomic nerve fibers



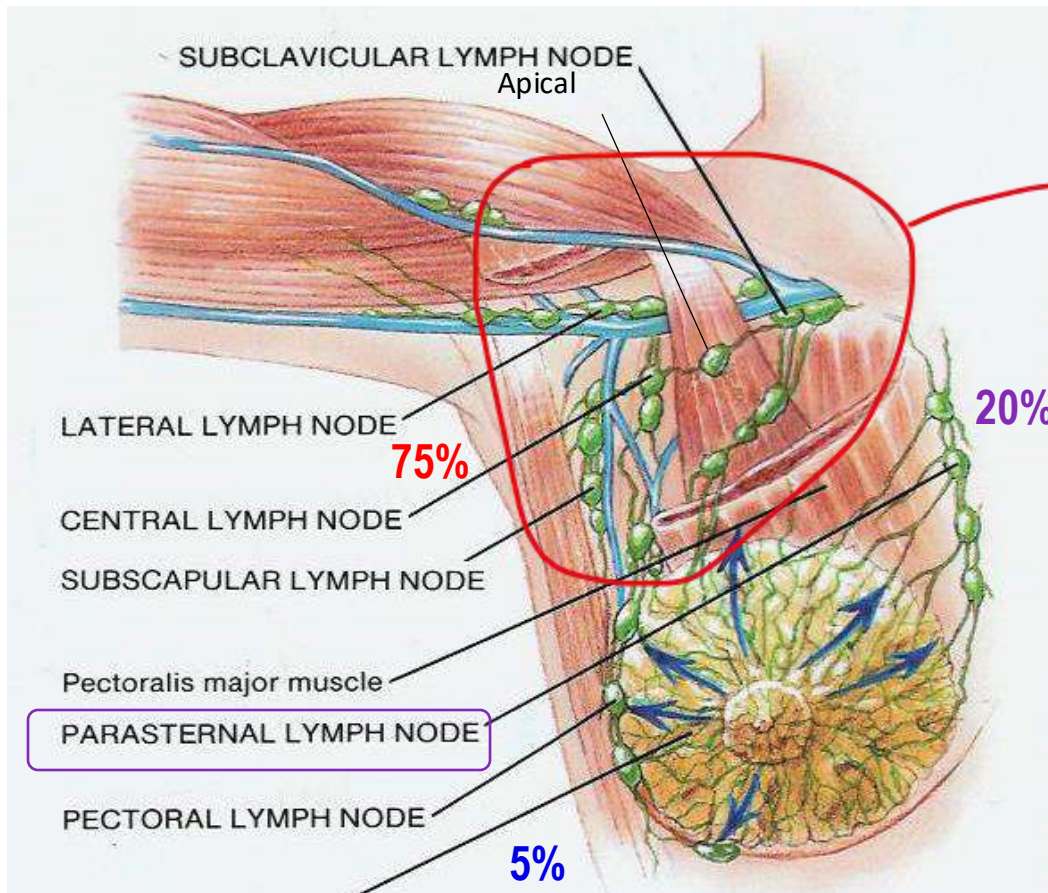
Lymph Drainage Of The Breast

Generally:

75% of the total lymph from the breast drains into → **Axillary lymph nodes**.

20% The **medial** quadrants drain into → **Internal thoracic LN (Parasternal)**

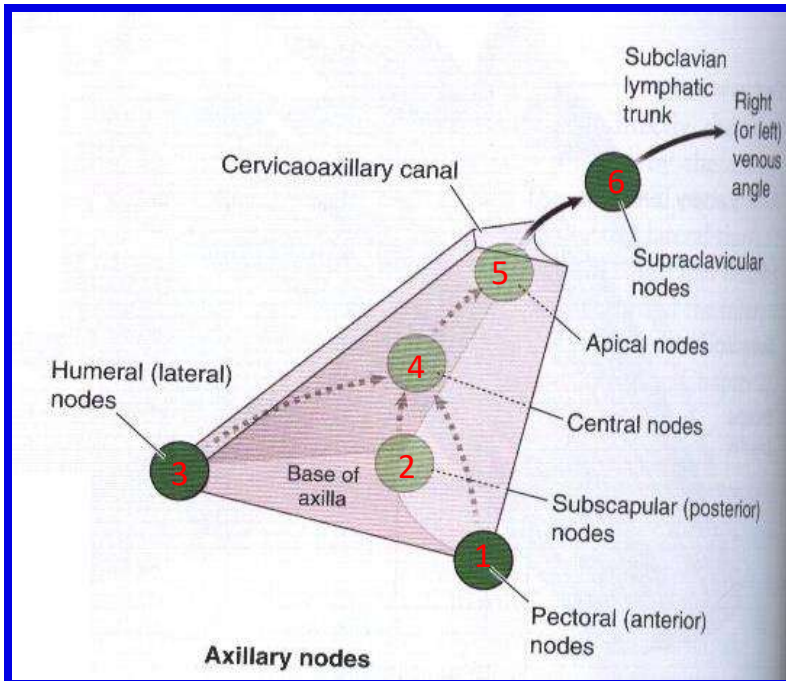
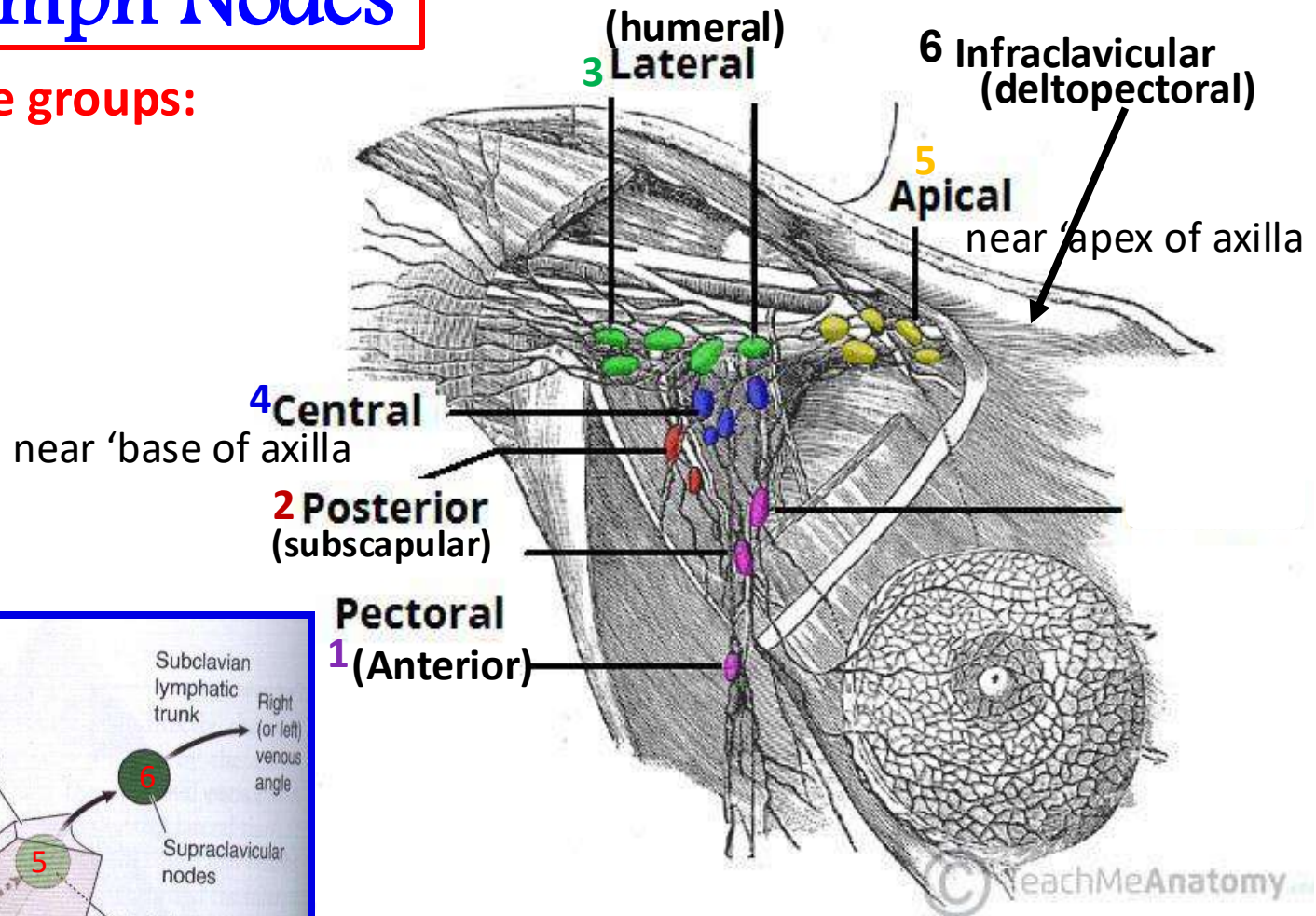
5% Few lymph vessels drain **posteriorly** into → **posterior intercostal LN**
→ to mediastinal LN.



What are the
axillary LN

Axillary Lymph Nodes

Arranged in five groups:



In summary:

Groups 1) 2) 3) drain into → 4
Then 4 drains into → 5 → 6 →
Subclavian lymph trunk

Lymph drainage of the breast

A-Nipple & areola : are drained by subareolar plexus to **pectoral & apical groups of axillary lymph nodes** .

B- Subcutaneous tissue :

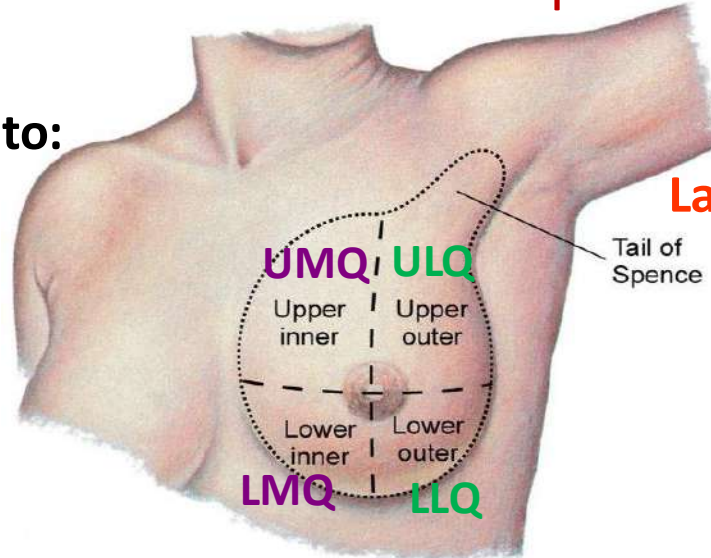
The breast is divided into 4 quadrants :

20%

Medial quadrants drain into:

UMQ :
Internal thoracic
(Parasternal) LN

LMQ:
Mediastinal LN



5%

Few lymph vessels drain **posteriorly** into
→ **posterior intercostal LN**

75%

Lateral quadrants drain into :

ULQ:
Pectoral (anterior) AxLN

LLQ + Axillary tail:
Subscapular (posterior) AxLN

☐ Some vessels communicate with the lymph vessels of:

- ✓ The opposite breast
- ✓ The anterior abdominal wall (subdiaphragmatic LN).

What is the clinical importance of this?

Lymphatic Drainage of the Upper Limb



Lymph Drainage of the Upper Limb

A) Superficial lymphatic:

Follows the superficial veins of the UL:

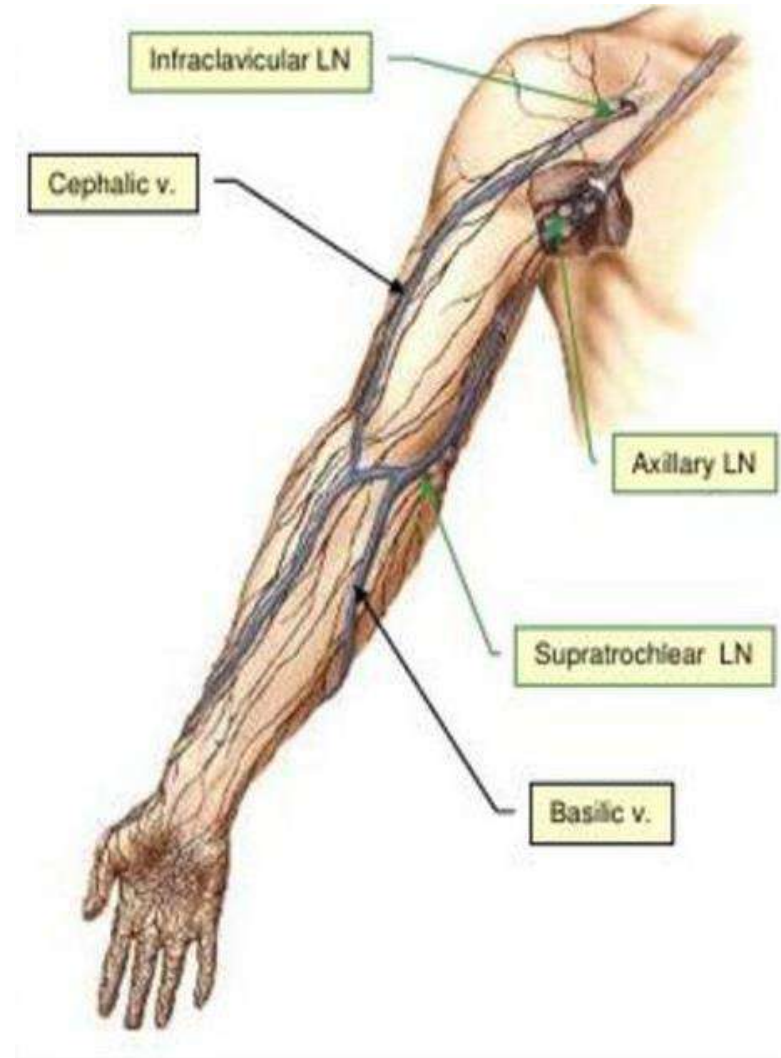
1-from the thumb , index and **lateral** side of the hand follows the **Cephalic vein** → to the **deltopectoral (infraclavicular) axillary LN**.

2- from the **medial** side of the hand follows the **Basilic vein** → to the **supratrochlear LN**, then to the **lateral (humeral) axillary LN**.

B) Deep Lymphatic:

Follows the arteries of the UL:

Drains into → **lateral (humeral) group**.

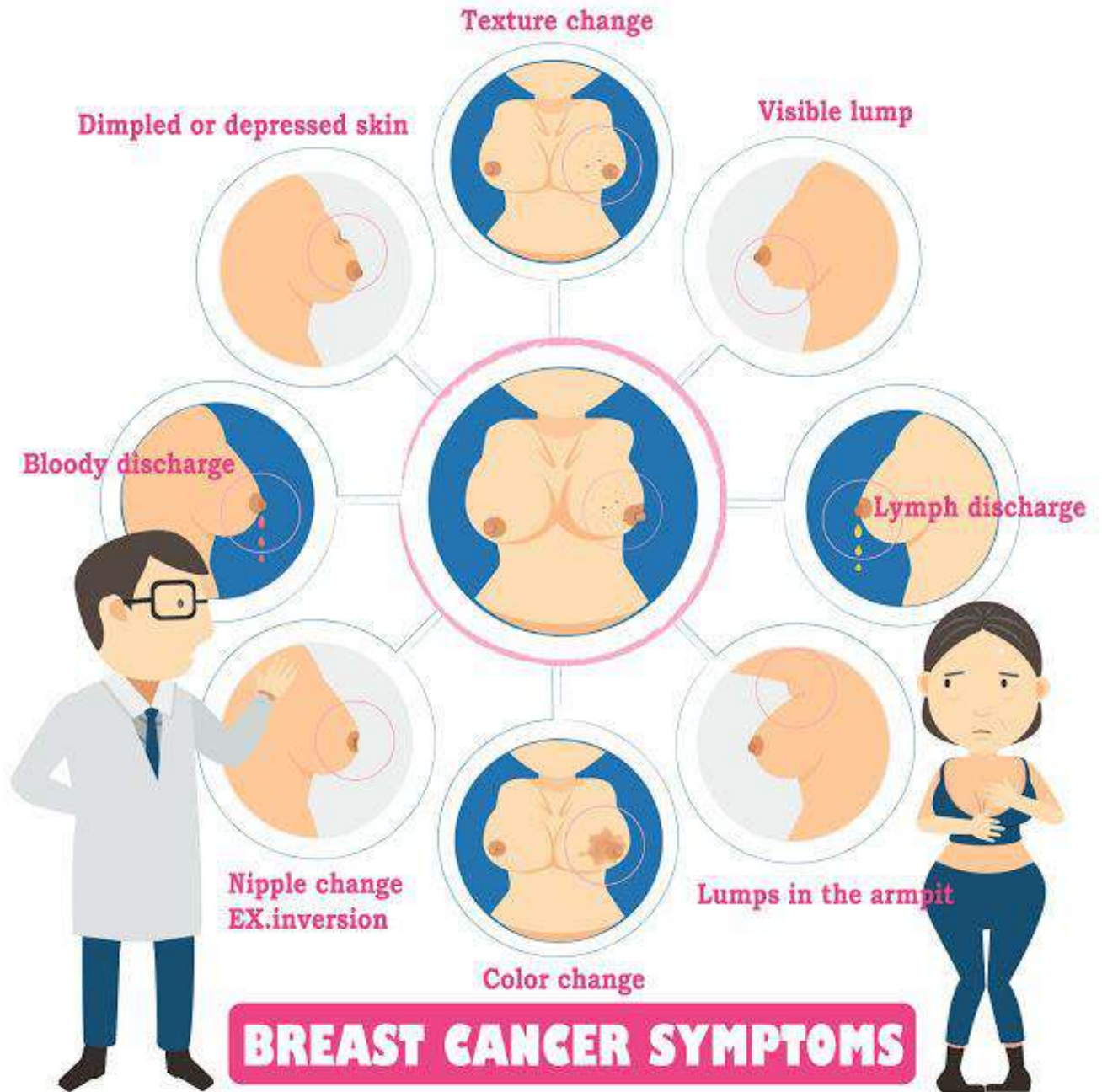


Clinical Notes



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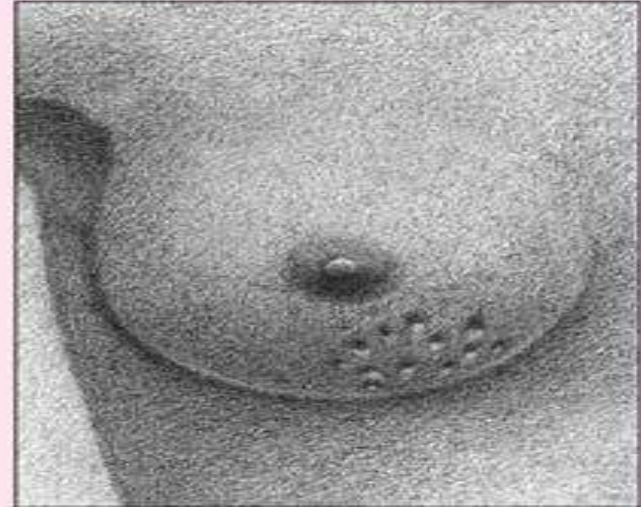
The best
protection
is early
detection



Breast Carcinoma

- 60% of breast carcinoma occurs in the *Upper Lateral Quadrant*.
- Firm, immobile **mass**
- The fibrous septa (suspensory ligament of Cooper) may be **contracted** → leading to **pitting** of the overlying skin.
- Invasion of the underlying deep fascia → **limited mobility** of the breast.
- Pulling on the lactiferous ducts → **Retracted nipple**

In peau d'orange, the skin appears to be pitted (as shown below). This condition usually indicates late-stage breast cancer.



❖ If breast carcinoma is suspected, the following structures should be examined:

- Axillary LN of the same side
- Axillary LN and breast of the opposite side (because anastomosing lymphatics)

In case of localized cancer:

Treatment: simple mastectomy followed by radiotherapy to axillary LN.

Breast abscess

1. A breast abscess may occur during lactation due to entrance of bacteria to breast tissue through a crack in nipple.
2. Because of the presence of the fibrous septa, the infection remains localized to one compartment or lobe to begin with.

Drainage of the breast abscess

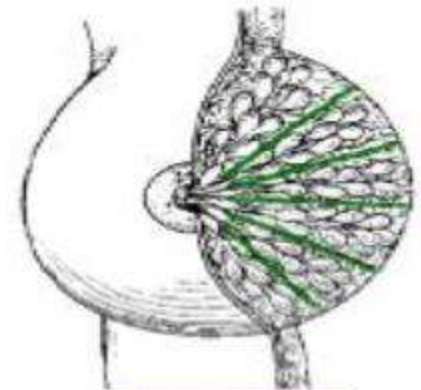
Abscesses should be drained through



Breast abscess



Breast abscess drainage



Breast lobes



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